STUDY ON AVERAGE SLEEP HOURS AND ITS DEPENDENCE ON VARIOUS ASPECTS

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Introduction:

Background

On average, Americans sleep around 7 hours a day. However, this might vary by region, gender, or age group. The article "Among teens, sleep deprivation an epidemic" (Richter) shows that lack of sleep in teenagers leads to a higher risk of negative mental consequences. However, another article shows as age increases, individuals sleep less and less compared to younger ages. In the "Sleep and Human Aging" (Mander) article, it is observed that older adults (>61 years) do not sleep as well as younger adults tend to sleep. To know more about which observation could be seen as correct, we decided to do a study on the relationship between age group and sleep time, also including other variables like Type.of.Days and Sex.

Dataset: How Much Sleep Do Americans Really Get? | Kaggle
This data set comes from the American Time Use Survey collected by the U.S. Bureau of Labor Statistics. To be specific, this data set is focusing on how much Americans sleep by gender and age. Include 945 data points with 6 distinct variables: year, average hours of sleep per day, standard error, type of day, age group, and sex.

How to Write a Statistical Report (with Pictures) - wikiHow

Context: Major Events in American History in the 21st Century (historycentral.com)

Hypothesis Test 1:

H0: There is a not significant difference in average hours of sleep among all age groups
H1: There is a significant difference in average hours of sleep among all age groups

Hypothesis Test 2:

H0: older age group has the highest average hours of sleep
H1: older age group do not have the highest average hours of sleep

Methods and Materials used:

1. **Data Visualization plots (with subset age groups of 15 to 24 years and 65 years and above):**
   - **Piechart**
   - **Double bar plots**

- Bar plots
    - **Bar plots**
    - **Stacked bar plots**
    - **Scatter/jitter plots**

2. **Hypothesis 1 testing using:**
    - **Pairwise t.test (Bonferroni):**
      Pairwise T-test is used to test if each age group within our data set has significantly different average hours of sleep.
    - **FDR:** Compares the multiple hypothesis of two different age groups having the same average hours of sleep.

3. **Hypothesis 2 testing using:**
   "When asked when they plan to retire, most people say between 65 and 67" (2)
    - **Tukey HSD tests: (**shows which sleeps more or less)**
      "Tukey's Honest Significant Difference (HSD) test is a post hoc test that we use to understand the significance of differences between pairs of age group means. The test is a follow up to Pairwise T-test after we find the significant difference between some of the age groups". Tukey test performed comparisons between each age group, obtaining the difference between Average Hours of Sleep. From this, we are able to determine the age group with the highest average hours of sleep. We used this result to determine which Age Group to conduct further tests on.

4. **Prediction Using:**
    - **All possible Regression:**
      All possible Regressions is used for model selection of the best linear model. The AIC is used to determine this. The AIC is a relative measure of prediction error, the lower the value the better the model is in comparison to the other models.
    - **Tree and Random Forest (Important variables):**
        - **Tree:** In a Decision tree, the root Node (topmost) implies the best predictor (independent variable). The Decision / Internal Node are the nodes with predictors (independent variables) being tested and each branch represents an outcome of the test. The Terminal Node holds a class label (category) with Final Classification Outcome. As tree usually tends to do overfitting.
            - <u>Pruning method to Correct Overfitting:</u> We check the number of leaves in the tree (i.e. size of the tree) and the error rate of the tree (i.e. misclassification rate or Sum of Squared Error).

            - The Complexity Parameter (Cp) of the true value of the smallest tree that has the smallest cross validation error. In regression, this means that the overall R-squared must increase by cp at each step.

    - **Random Forest regression:** It provides two outputs, means square error (MSE) and Node purity which is how well a predictor decreases variance. As MSE is a more reliable measure of variable importance, we tend to look at it more to get the optimum importance of variables.

        - **Important variables**: Use various linear regression models to get which variables have higher importance and so tend to affect the regression model

- **Regression Model Prediction (Linear and Interaction terms):**
    - **RMSE:** Lower values of RMSE indicate better fit. RMSE is a good measure of how accurately the model predicts the response.

    - **Error function:** Created a error function calculating mae, maps, rmse, mse.

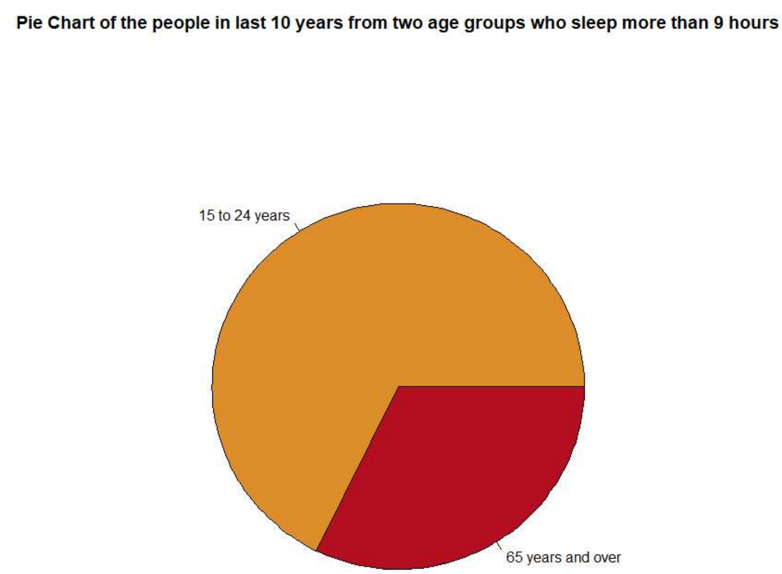
- **Bayesian Regression Model Prediction (Linear and Interaction terms):**
    - Based on the best model for prediction Average hours of Sleep of BIC value, and using does Markov chain Monte Carlo (MCMC) sampling with each age groups
    - The result shows the average of 20000 predictions of average hours of sleep in each age group.

Analysis and Results:

"Sleep loss was most pronounced in the youngest group (ages 24-32)" (1)

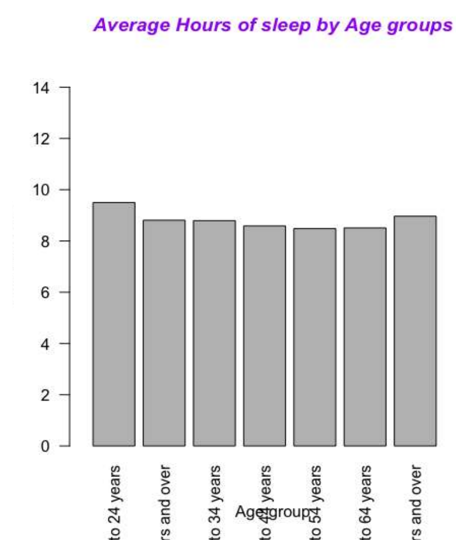
a. **Data Visualization plots:**

i. **Pie Charts:**
Using a pie chart of people in the last 10 years from two age groups who sleep more than 9 hours, we found that the 15 to 24 years old sleep more by a much greater proportion than the 65 years and older age group.

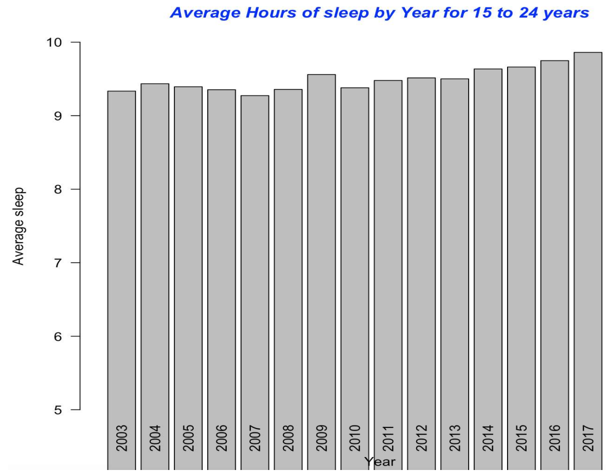


ii. **Bar plots:**

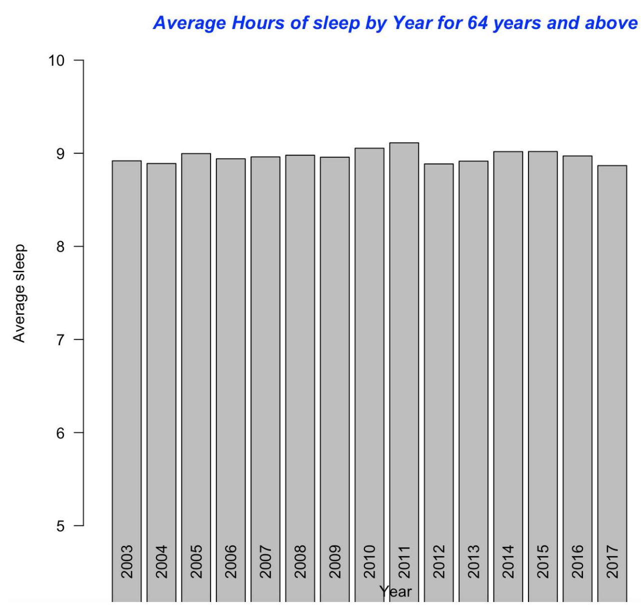
Bar plot for <u>Average hours of sleep per group</u> shows that the highest number of average sleep hours is for the 15 to 24 age group. After 15 to 24 years old, the 65 years and over age group have the second highest average sleeping hours per day. After the 15 to 24 years old group, the average sleep hours decreases as the age group increase from 25 to 64 years. Hence, we choose to look at the two highest sleeping age groups, 15 to 24 years and 65 years and over.



Bar plot for the <u>15 to 24 years age group over years</u> from 2003-17 shows a general trend of increasing average sleeping hours as the years increase. However, there is quite a sudden decrease in average sleep hours in 2005-7 and there is a peak increase in 2009 with a drastic decrease in sleep hours in 2010. *Add context*

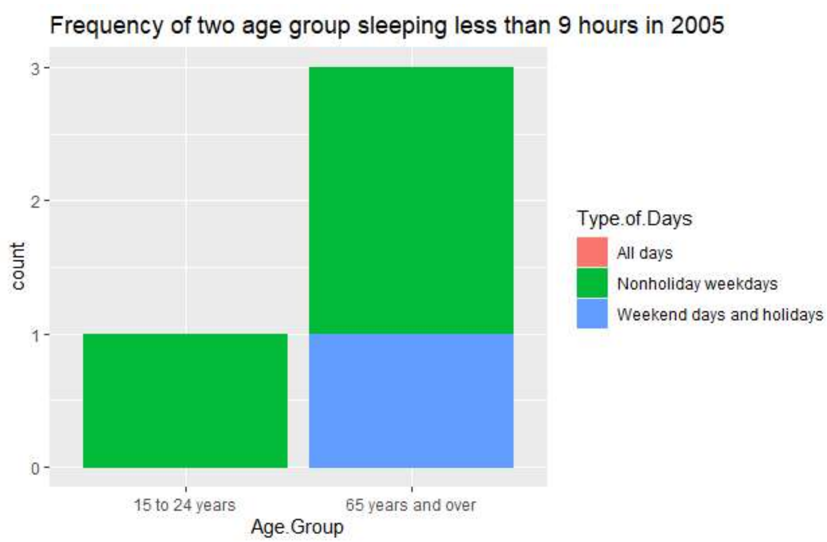


Bar plot for <u>65 years and over age group from 2003-17</u> shows a general trend of quite stable average hours of sleep over the years (about 8.9). Though there is an increase in average sleep hours till 2011 (about 9), there is a sudden drop in the average sleeping hours trend. Especially, 2012 and 2013 (maybe due to shooting and bombing stress *see relative events*) show the sudden drop after which the average sleeping hours begin to rise till 2015. The sudden decrease trend from 2015 to 2017 could be due to Trump's win in the 2016 election.

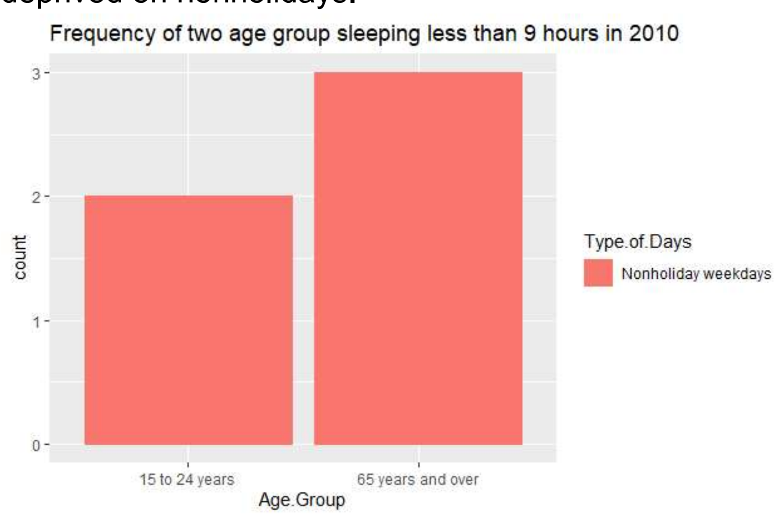


iii. **Stacked bar plot:**

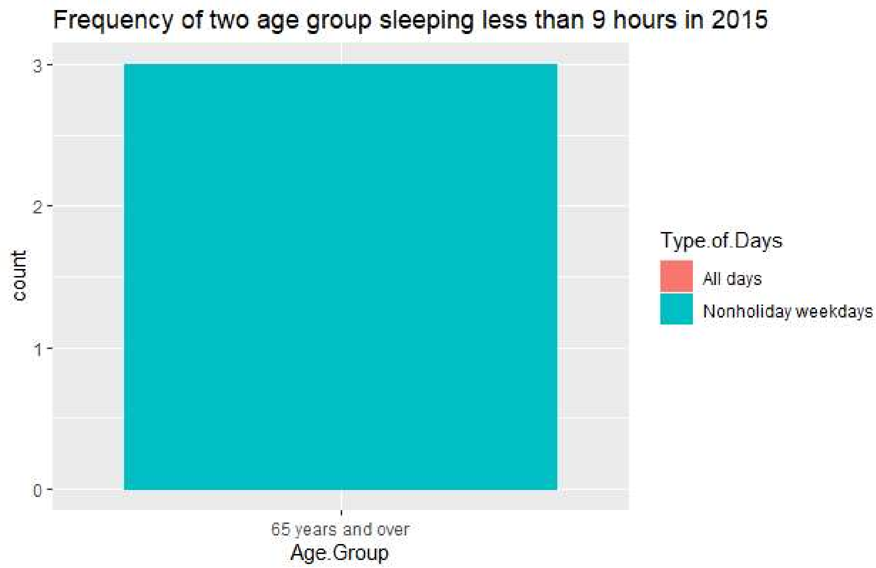
Frequency of two age groups sleeping less than 9 hours in 2005 shows that about 1 person in the 15 to 24 years age group sleeps less than 9 hours only on nonholiday weekdays. However, about 1 person in the 65 years and above group sleeps less than 9 hours only on weekend days and holidays, but about 3 people in the same group sleep less than 9 hours only on nonholiday weekdays. This bar plot indicates that not many in the 15 to 24 years age group are sleep deprived (only when nonholidays, which could be due to an increase in work and high-stress levels), and more people 65 years and over are sleep deprived in nonholidays as well as on holidays.



The frequency of two age group sleeping less than 9 hours in 2010 shows that about 2 people in the 15 to 24 years age group sleep less than 9 hours only on nonholiday weekdays and about 3 people in the 65 years and above group sleep less than 9 hours only on nonholidays. This bar plot indicates that there is an increase in people in the 15 to 24 years age group who are sleep deprived (on nonholidays, it could be due to social media and competitive work employment leading to higher stress levels). Compared to the stacked bar plot for 2005, about the same number of people 65 years and over are sleep deprived on nonholidays.

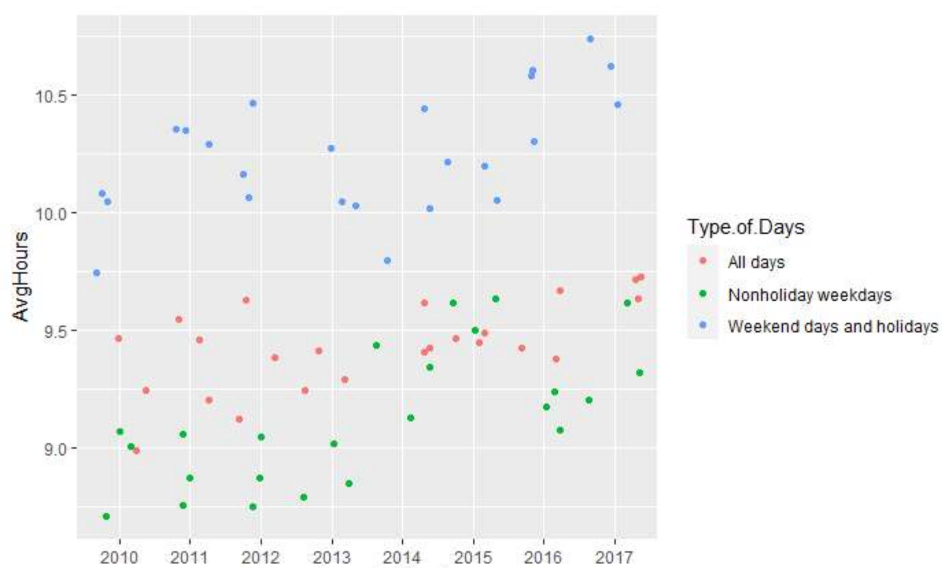


Frequency of two age groups sleeping less than 9 hours in 2015 shows that only people who sleep less than 9 hours are 65 years and over and that they only have sleep deprivation on nonholiday weekdays. Surprisingly, the age group 15 to 24 years includes everyone who sleeps more than 9 hours which shows the increase in average sleep hours in the age group over the past 5 years.

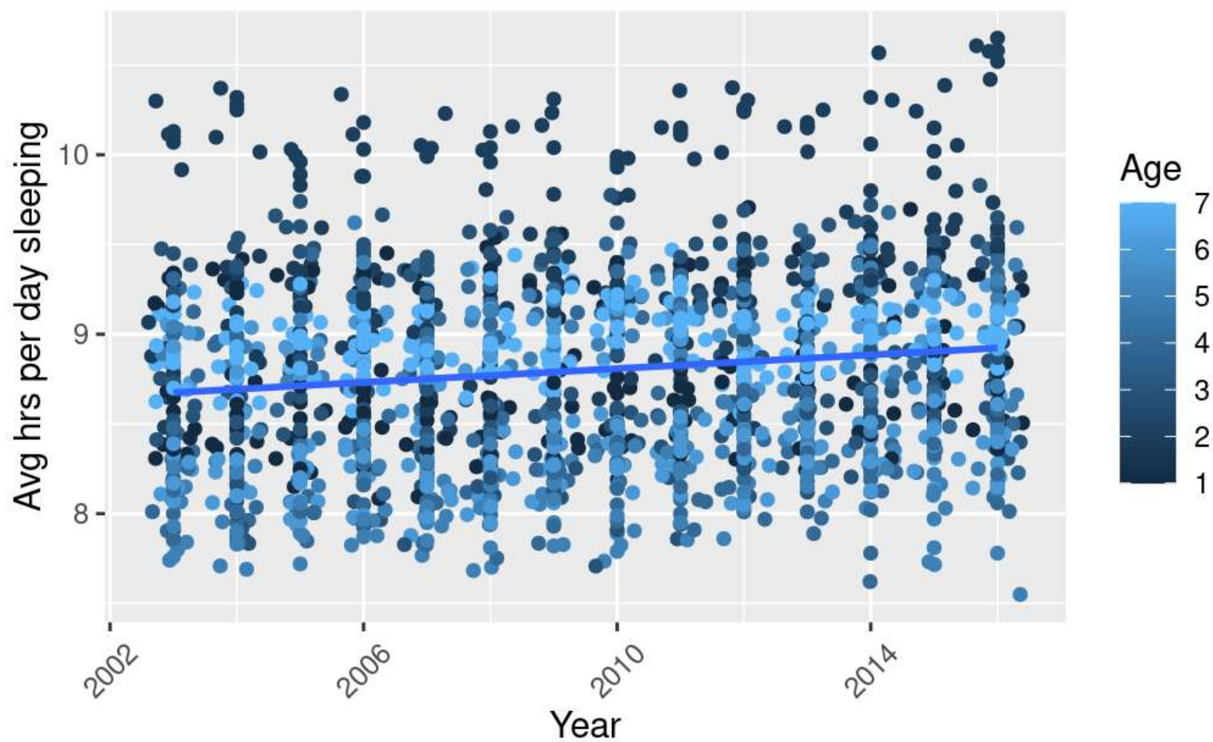


iv. **Scatter/Jitter plot:**

Jitter plot to look at various age groups' average sleep hours over years group by type of days: Compare the non-holiday weekdays and the weekend days and holidays with all day, people tend to sleep less on nonholiday weekdays and sleep more in weekend days and holidays.



Jitter plot to look at various age groups' average sleep hours over years:
More people below 24 years sleep more than 10 hours compared to those 35 and above years that tend to sleep about 7.5 to 9.5 average years. Also, the lightest color (55 and above) tend to sleep mostly between 9 to 9.5 average hours in a day. And we can see in general, sleeping time increase as the year increase.



b. Hypothesis testing results analysis:

1. **Hypothesis test 1:** *Null hypothesis is that each age group has the same average sleep time (difference equal to 0), and our alt hypothesis is that each age group has a different average sleep time (difference not equal to 0).*
    a. Pairwise t-test shows us that there while there is a significant difference between a majority of, there is an exception. Age groups 15 years and below and 25 to 34 years shows that there is not a significant difference in average hours of sleep. Similarly, the age groups 55 to 64 years and 45 to 54 years are also not significantly different from one another.

2. **Hypothesis test 2:** *Null hypothesis is that the elder age group (>65) has the greatest average sleep time than other age groups, and our hypothesis is that*

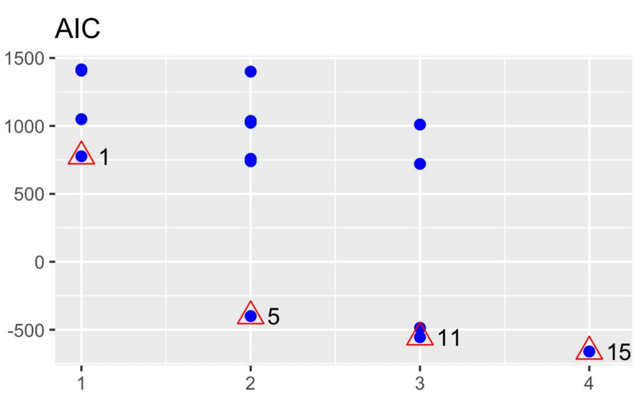
the elder age group (>65) does not have the greatest average sleep time among other age groups. We get all of the F values as significant and reject H0, so all of the means are different. Hence, we conclude that each age group has a different average sleep time. Now we used

a. <u>TukeyHSD test:</u> to assess the significance of differences between the pairs of age groups' means. This is a follow-up after using ANOVA where the F values show the significantly differing pairs of age groups. We found that the most significantly different pairs of age groups were 45 to 54 years and 15 to 24 years. 15-24 years tend to sleep more than 45-54 years by 1.01518519. Though not the highest significant difference among other pairs, the 65 years and older age group had significantly different means with the rest of the age groups, showing that they slept on average more hours. The exception to this would be 15-24 years, which tends to sleep more than 65 years and over by 0.53340741. Other age groups like 15 years and below, 25 to 24 years, 35 to 44 years, and 55 to 64 years old (mostly all age groups) significantly differed in means than the 65 years and over. Hence, we reject H0 and concluded that the elder age group (>65) does not have the greatest average sleep time among other age groups.

c. Model Selection and Prediction:

1. **<u>Best Linear Regression Model and Accuracy check:</u>**

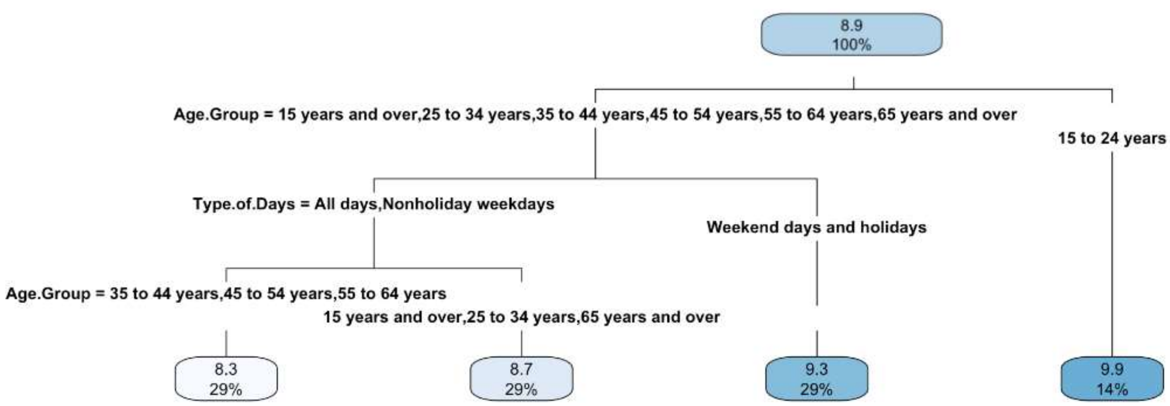
   a. <u>All possible Regression:</u> We used all possible regression on a full regression model to test all possible subsets and potential independent variables to include. From this we obtained 4 models with significant independent variables. Looking at the AIC of the 4 models to determine model selection, we saw that the model including the variables Year, Age Group, Type of Days and Sex as regressors is the best.



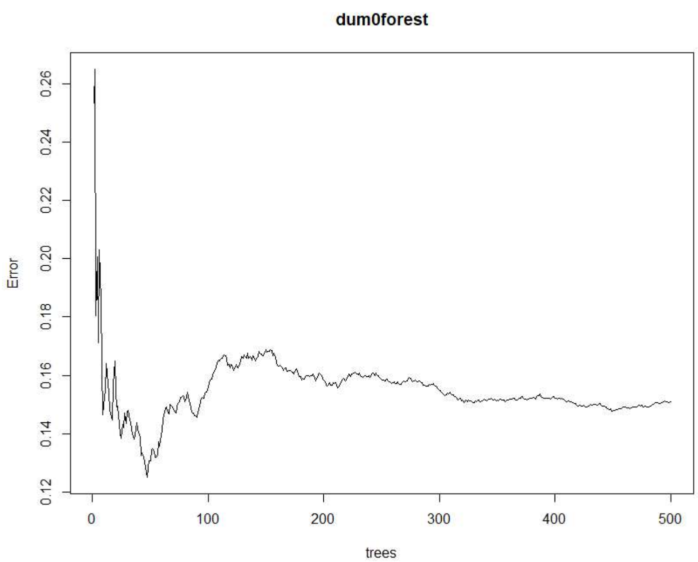
Best Subsets Regression

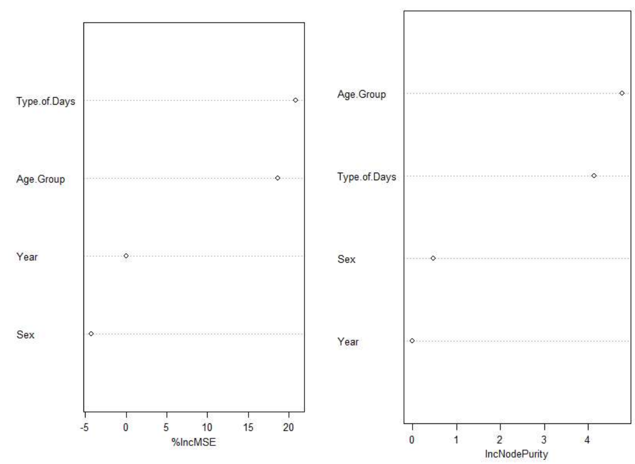
Model Index	Predictors
1	Type.of.Days
2	Age.Group Type.of.Days
3	Year Age.Group Type.of.Days
4	Year Age.Group Type.of.Days Sex

b. Trees: From our decision tree we found that the average sleep hours per day for all age groups (root node) is 8.9. Among all age groups, the 15 to 24 years age group has average sleep hours greater than 8.9, about 9.9 (constituting about 14% of the total population from all age groups). The age groups with average sleep hours less than 8.9 are 15 and below, 25 to 24 years, 35 to 44 years, 45 to 55 years, 54 to 65 years, and 65 years and over. Among the age groups that sleep less than 8.9 hours per day, the people who sleep <8.9 hours on weekend days and holidays have about 8.7 average hours of sleep per day (constituting about 29% of the total population from all age groups). The people who sleep <8.9 hours on all days and non-holiday weekdays and are in the 35 to 44 years, 45 to 55 years, and 55 to 65 years age groups tend to sleep about 8.3 average hours per day (constituting about 29% of the total population from all age groups). The people who sleep <8.9 hours on all days and non-holiday weekdays and are 15 and below, 25 to 34 years, and 65 years and over age groups sleep about 8.7 average hours per day, making them 29% of the total population from all age groups.



b. RandomForest: The higher %IncMSE we have, the higher there is decrease in model accuracy and the higher importance will be of that certain variable. We found that the highest %IncMSE is 18.988181 for Type.of.Days and second highest is 16.546491 for Age Groups. As for the model accuracy, MSE is a more reliable measure of variable importance, and we tend to look at it more to get the optimum importance of variables. Thus, by %IncMSE we conclude that the most important variables are Type.of.Days and Age.Group and they will mainly affect the model accuracy which would mean better prediction accuracy.





2. <u>Best Regression Model:</u>
    a. Linear model using Interaction terms: By using the response variable as Average sleep hours per day, we use various predictors' combinations and even add interaction terms in order to find the best model for prediction. First, we only try to find the best linear regression model using trees, RandomForest to see how
    b. AIC & BIC: The AIC and BIC appeared lowest for model1

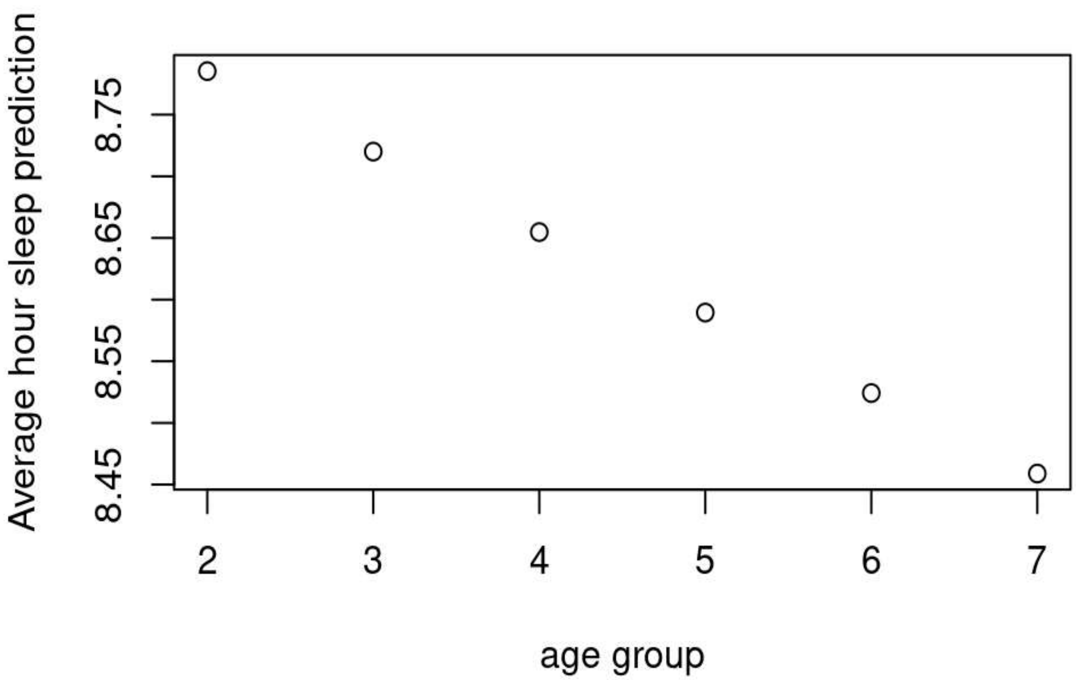
3. <u>Prediction for Average Sleep hours in 2017 (train):</u>
    a. RMSE: model1 is seen to have the lowest RMSE

d. Bayesian Prediction:

1. <u>Build the model and Simulation via rstanarm</u>
    a. Model Selected: Using the model: (`Avg hrs per day sleeping` ~ Gender + Year + Age + Age:Gender + Day:Gender + Year:Age + Year:Day) which is based on the BIC result
    b. Family: We are using the "gaussian" because we are assum that the `Avg hrs per day sleeping` distribute normally.
    c. Check Model: MCMC trace shows if chains mixed well; MCMC dens overlay shows if the chains have sufficient length; MCMC acf shows if the variable is autocorrelated; If Sample size ratio is

greater than 10%, it shows that there are enough data set; Rhat is close to 1 smaller than 1.05 shows the stimulation is stable.

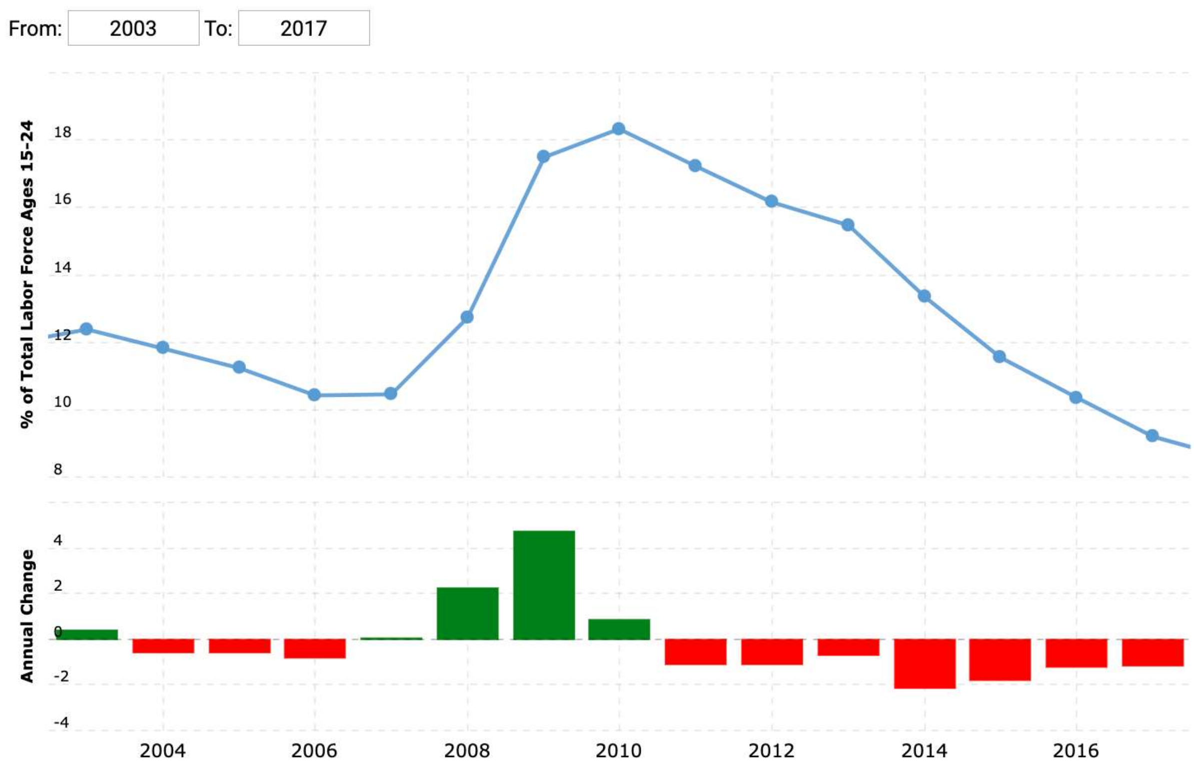
2. <u>Posterior prediction</u>
    a. Function: the posterior prediction function gives 20000 predictions based on the existing data; and we are only print out the average of those predictions with each age groups.
    b. Result: From the graph, we can see that as the age increase, the average hour of sleep decrease



Relative events:

2005→ Hurricane Katrina
2007→ iphone released to public
2008→ Election (Obama wins)
2012→ Sand Hook shooting, Obama reelection
2013→ Boston marathon bombing
2016→ Election (Trump wins)

Unemployment rate of age group 15-24 years old from 2003-2017:



Conclusion:

Starting with our focus on looking at main four variables: Year, Type of Days, and Age Groups. Then looking at the relationship of Age groups, we find that all age groups were different than each other (differences in their means were significantly different). After finding what age group differ, we use the tukey test which we cannot concluded that the elder age group (>65) does have the greatest average sleep time among other age groups.Then we do the tree, where the people who sleep <8.9 hours on all days and non-holiday weekdays and are 15 and below, 25 to 34 years, and 65 years and over age groups sleep about 8.7 average hours per day, making them 29% of the total population from all age groups. Then, by %IncMSE we conclude that the most important variables are Type.of.Days and Age.Group and they will mainly affect the model accuracy which would mean better prediction accuracy. Then for the model selection, model1 is seen to have the lowest RMSE, BIC and AIC. Then we use the best model to do the Bayesian prediction on average sleep hour in 2017 for different age group shows that as the age increase, the predict average sleep hour decrease, which is support for that elder people sleep less.

SOURCES:

1 [The Impact of the 2016 US General Election on Adult Sleep in Seattle, WA (washington.edu)]

2 [What is the optimal age to retire? | Fifth Third Bank (53.com)]

3 https://www.macrotrends.net/countries/USA/united-states/youth-unemployment-rate

Appendix:

```
library(tree)
library(rpart)
library(randomForest)
library(ISLR)
library(pROC)
library(gam)
library(olsrr)
Sleep <- read.csv("Downloads/TimeSleeping.csv",
    col_types = cols(index = col_skip(),
        Period = col_skip(), Activity = col_skip()),
           colClasses = c("numeric", "factor", "factor", "numeric", "numeric", "factor",
           "factor","factor","factor"))
View(Sleep)
names(Sleep)[names(Sleep) == "Avg.hrs.per.day.sleeping"] <- "AvgHours"
head(Sleep)

#Create new columns
Sleep$Day <- with(Sleep, ifelse(Type.of.Days == 'All days', '1',
                        ifelse(Type.of.Days == 'Nonholiday weekdays', '2', '3')))
Sleep$Age <- with(Sleep, ifelse(Age.Group == '15 years and over', '1',
                         ifelse(Age.Group == '15 to 24 years', '2',
                         ifelse(Age.Group == '25 to 34 years', '3',
                         ifelse(Age.Group == '35 to 44 years', '4',
                         ifelse(Age.Group == '45 to 54 years', '5',
                         ifelse(Age.Group == '55 to 64 years', '6', '7'
)))))))
Sleep$Gender <- with(Sleep, ifelse(Sex == 'Men', '0',
                         ifelse(Sex == 'Women', '1', '2')))
#Change the data to numeric
Sleep$Year <- as.numeric(Sleep$Year)
Sleep$Day <- as.numeric(Sleep$Day)
Sleep$Age <- as.numeric(Sleep$Age)
Sleep$Gender <- as.numeric(Sleep$Gender)
barplot(Sleephravg, main="Average Hours of sleep by Age groups", font.main=4,
col.main="purple", xlim=c(0,9), ylim=c(0, 15), xlab="Age group", ylab="Average sleep", las=2)
```

```
barplot(SleepYearsyoung, main="Average Hours of sleep by Year for 15 to 24 years",
font.main=4, col.main="blue", xlim=c(0,20), ylim=c(5, 10), xlab="Year", ylab="Average sleep",
las=2)
barplot(SleepYearsold, main="Average Hours of sleep by Year for 64 years and above",
font.main=4, col.main="blue", xlim=c(0,20), ylim=c(5, 10), xlab="Year", ylab="Average sleep",
las=2)
S1 <-  Sleep %>%
   filter((Age.Group=="15 to 24 years"| Age.Group=="65 years and over") & (AvgHours>9)&(Year>2009))
table(S1$Age.Group)
slices <- c(63, 30)
lbl<- c("15 to 24 years", "65 years and over")
pie(slices, labels = lbl, main="Pie Chart of the people in last 10 years from two age groups who sleep
more than 9 hours", col = wes_palette("FantasticFox1", 2, type = "continuous"))
S5 <- Sleep %>%
  filter((Age.Group=="15 to 24 years"| Age.Group=="65 years and over") & AvgHours<9 &Year==2005)

ggplot(S5, aes(x=Age.Group, fill=Type.of.Days, color=Type.of.Days))+
  geom_bar(position="identity")+ ggtitle("Frequency of two age group sleeping less than 9 hours in
2005")
S6 <- Sleep %>%
  filter((Age.Group=="15 to 24 years"| Age.Group=="65 years and over") & AvgHours<9 &Year==2010)

ggplot(S6, aes(x=Age.Group, fill=Type.of.Days, color=Type.of.Days))+
  geom_bar(position="identity")+ ggtitle("Frequency of two age group sleeping less than 9 hours in
2010")
S5 <- Sleep %>%
  filter((Age.Group=="15 to 24 years") &Year>2009)

ggplot(S5, aes(x = Year, y = AvgHours, color = Type.of.Days)) +
  geom_jitter(height = 0.25)+  theme(axis.text.x = element_text(angle = 45, vjust = 0.5)
#Differs from scatter plot and makes points for each year better to see
ggplot(S5, aes(x = as.character(Year), y = AvgHours, color = Type.of.Days)) +
  geom_point()

pairwise.t.test(Sleep$AvgHours, Sleep$Age.Group, p.adjust.method = "bonferroni", paired =
TRUE)

#     Pairwise comparisons using paired t tests

#data:  Sleep$AvgHours and Sleep$Age.Group

#                   15 to 24 years 15 years and over 25 to 34 years
#15 years and over < 2e-16        -                 -
#25 to 34 years    < 2e-16        1.00000           -
#35 to 44 years    < 2e-16        < 2e-16           < 2e-16
```

```
#45 to 54 years    < 2e-16        < 2e-16            < 2e-16
#55 to 64 years    < 2e-16        < 2e-16            < 2e-16
#65 years and over < 2e-16        3.2e-07            7.1e-05
#                  35 to 44 years 45 to 54 years 55 to 64 years
#15 years and over -              -              -
#25 to 34 years    -              -              -
#35 to 44 years    -              -              -
#45 to 54 years    4.9e-13        -              -
#55 to 64 years    0.00041        1.00000        -
#65 years and over < 2e-16        < 2e-16        < 2e-16

#P value adjustment method: bonferroni
```

```
library(dplyr)

Sleep1<-Sleep%>%
  filter(!is.na(Sleep$AvgHours)&!is.na(Sleep$Age.Group))
subyounger<-subset(Sleep, Sleep$Age.Group=="15 to 24 years")
subolder<-subset(Sleep, Sleep$Age.Group=="65 years and over")

#Checking which age group has highest mean
Sleephravg<-tapply(Sleep$AvgHours,Sleep$Age.Group,mean)
SleepYearsyoung<-tapply(subyounger$AvgHours,subyounger$Year,mean)
SleepYearsold<-tapply(subolder$AvgHours,subolder$Year,mean)

#Compare the difference of mean between each significant age groups
fit_AvgSleep<-aov(Sleep$AvgHours~Sleep$Age.Group)
TukeyHSD(fit_AvgSleep)
```

#	diff	lwr	upr	p adj
#15 years and over-15 to 24 years	-0.68896296	-0.846348933	-0.53157699	0.0000000
#25 to 34 years-15 to 24 years	-0.70614815	-0.863534118	-0.54876218	0.0000000
#35 to 44 years-15 to 24 years	-0.91303704	-1.070423007	-0.75565107	0.0000000
#45 to 54 years-15 to 24 years	-1.01518519	-1.172571155	-0.85779922	0.0000000
#55 to 64 years-15 to 24 years	-0.99000000	-1.147385970	-0.83261403	0.0000000
#65 years and over-15 to 24 years	-0.53340741	-0.690793377	-0.37602144	0.0000000
#25 to 34 years-15 years and over	-0.01718519	-0.174571155	0.14020078	0.9999094
#35 to 44 years-15 years and over	-0.22407407	-0.381460044	-0.06668810	0.0005636
#45 to 54 years-15 years and over	-0.32622222	-0.483608192	-0.16883625	0.0000000
#55 to 64 years-15 years and over	-0.30103704	-0.458423007	-0.14365107	0.0000004
#65 years and over-15 years and over	0.15555556	-0.001830414	0.31294153	0.0551174
#35 to 44 years-25 to 34 years	-0.20688889	-0.364274859	-0.04950292	0.0021102
#45 to 54 years-25 to 34 years	-0.30903704	-0.466423007	-0.15165107	0.0000002
#55 to 64 years-25 to 34 years	-0.28385185	-0.441237822	-0.12646588	0.0000026

#65 years and over-25 to 34 years	0.17274074	0.015354771	0.33012671	0.0208485
#45 to 54 years-35 to 44 years	-0.10214815	-0.259534118	0.05523782	0.4689602
#55 to 64 years-35 to 44 years	-0.07696296	-0.234348933	0.08042301	0.7771287
#65 years and over-35 to 44 years	0.37962963	0.222243660	0.53701560	0.0000000
#55 to 64 years-45 to 54 years	0.02518519	-0.132200785	0.18257116	0.9991687
#65 years and over-45 to 54 years	0.48177778	0.324391808	0.63916375	0.0000000
#65 years and over-55 to 64 years	0.45659259	0.299206623	0.61397856	0.0000000

 # As 15-24 years tend to sleep more than 45-54 years by 1.01518519. Hence, we reject H0 and conclude that 45-54 years old sleep less than the 15-24 years.
#So, the average sleep of 65 years is greater than that of other groups except the group of 15 to 24 years

#Errorfunction.R (source)

```
#Custom error function
regr.error <- function(predicted,actual){
  #MAE
  mae <- mean(abs(actual-predicted))
  #MSE
  mse <- mean((actual-predicted)^2)
  #RMSE
  rmse <- sqrt(mean((actual-predicted)^2))
  #MAPE
  mape <- mean(abs((actual-predicted)/actual))
  errors <- c(mae,mse,rmse,mape)
  names(errors) <- c("mae","mse","rmse","mape")
  return(errors)
}
```

#Train and test dataset
#split your dataset
#split your dataset
#testing dataset

```
Sleep.test <- subset(Sleep, Sleep$Year!=2017)
```

#train dataset

```
Sleep.train<- subset(Sleep, Sleep$Year==2017)

rmsetest<-function(mod,datset){
model.predict<-predict(mod,Sleep.train,type="response")
predicted<-data.frame(model.predict)
 write.csv(predicted,file="SleepPred.csv")
 regr.error(model.predict,datset$AvgHours)
}

model<- lm(AvgHours ~ Age.Group+ Sex+ Type.of.Days + Year, data=Sleep.test)
```

rmsetest(model, Sleep.test)

```
#       mae         mse        rmse       mape
#0.62748992 0.59729319 0.77284746 0.07184462
```

model1<- lm(AvgHours ~ Sex + Year + Age.Group+ Age.Group:Sex + Type.of.Days:Sex + Year:Age.Group + Year:Type.of.Days, data=Sleep.test)
rmsetest(model1,Sleep.test)

```
#       mae         mse        rmse       mape
#0.62233249 0.59106679 0.76880868 0.07125918
```

model2<- lm(AvgHours ~ Sex + Year + Age.Group+ Age.Group:Sex + Type.of.Days:Sex + Year:Age.Group + Year:Type.of.Days +Year:Sex, data=Sleep.test)
rmsetest(model2,Sleep.test)

```
#       mae         mse        rmse       mape
#0.62375106 0.59354452 0.77041840 0.07142107
```

#---
#Prediction in different method (Saves predicted files with names)
model<- lm(AvgHours ~ Age.Group+ Sex+ Type.of.Days + Year, data=Sleep.test)
model.predict<- predict(model, Sleep.test)
Predicted<- data.frame(model.predict)
write.csv(Predicted,file="Avg.sleep.hours.predictions.csv")
regr.error(model.predict, Sleep.test$AvgHours)

model1<- lm(AvgHours ~ Sex + Year + Age.Group+ Age.Group:Sex + Type.of.Days:Sex + Year:Age.Group + Year:Type.of.Days, data=Sleep.test)
model.predict1<- predict(model1, Sleep.test)
Predicted1 <- data.frame(model.predict1)
write.csv(Predicted1,file="Avg.sleep.hours.predictions1.csv")
regr.error(model.predict1, Sleep.test$AvgHours)

model2<- lm(AvgHours ~ Sex + Year + Age.Group+ Age.Group:Sex + Type.of.Days:Sex + Year:Age.Group + Year:Type.of.Days +Year:Sex, data=Sleep.test)
model.predict2<- predict(model2, Sleep.test)
Predicted2 <- data.frame(model.predict2)
write.csv(Predicted2,file="Avg.sleep.hours.predictions2.csv")
regr.error(model.predict2, Sleep.test$AvgHours)

#Lowest rmse means better prediction values for Avg Sleep Hours. Model 2 is better looking at rmse.

#MODEL SELECTION/IMPRTANCE OF VARIABLES
#BIC:
> BIC(model)
[1] 1084.736
> BIC(model1)
[1] -520.6169
> BIC(model2)
[1] -510.056

#AIC
> AIC(model)
[1] 1036.914
> AIC(model1)
[1] -697.558
>AIC(model2)
[1] -696.5615

#all possible Regression using AIC selection:
> k<-ols_step_all_possible(model,sbc=TRUE)
> plot(k)
> k

```
library(tree)
library(rpart.plot)
library(rpart)
library(randomForest)
library(ISLR)
library(pROC)
library(gam)
dim(Sleep)
#[1] 945   9
dum0tree<-tree(AvgHours ~ Year + Age.Group + Type.of.Days +
Sex,data=Sleep.train)
dum0rpart<-rpart(AvgHours ~ Year + Age.Group + Type.of.Days +
Sex,data=Sleep.train)
rpart.plot(dum0rpart, type=3)
printcp(dum0rpart)#look at the rel error for other reg models to check
which model gives lowest
```

```
Regression tree:
rpart(formula = AvgHours ~ Year + Age.Group + Sex + Type.of.Days,
    data = Sleep.train)

Variables actually used in tree construction:
[1] Age.Group    Type.of.Days

Root node error: 21.526/63 = 0.34168

n= 63

        CP nsplit rel error  xerror     xstd
1 0.433779      0   1.00000 1.05565 0.203559
2 0.312417      1   0.56622 0.91374 0.185637
3 0.046301      2   0.25380 0.37115 0.070132
4 0.010000      3   0.20750 0.32197 0.071344
```

```
dum0forest<-randomForest(AvgHours ~ Year + Age.Group + Type.of.Days +
Sex,data=Sleep.train,importance=T, prox imity=T)
par(mfrow=c(1,1))
plot(dum0forest)
dum0forest
Test50_rf_pred <- predict(dum0forest, Sleep.test, type="class")
table(Test50_rf_pred, Sleep.train$Creditability)
importance(dum0forest)
varImpPlot(dum0forest,  main="", cex=0.8)
```

```
> importance(dum0forest)
                %IncMSE IncNodePurity
Year           0.000000     0.0000000
Age.Group     18.597100     4.7786621
Type.of.Days 20.843715     4.1274922
Sex           -4.218628     0.4663723
```

#Negative better than positive, so model 2 better is
#FDR
#pvalues obtained from pairwise t-test
#significance in average hours of sleep between age groups

vec<-c(2e-16, 2e-16, 0.17, 2e-16, 2e-16, 2e-16, 2e-16, 2e-16, 2e-16, 2.4e-14, 2e-16, 2e-16, 2e-16, 2.0e-05, 0.15, 2e-16, 1.5e-08, 3.4e-06, 2e-16, 2e-16, 2e-16)
fdr(vec,0.1,F)

[1] 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16 2.0e-16
[15] 2.0e-16 2.4e-14 1.5e-08 3.4e-06 2.0e-05
#[1] 1 2 4 5 6 7 8 9 11 12 13 16 19 20 21 10 17 18 14

#BAYESIAN PREDICTION:

```
Sleep <- stan_glm(`Avg hrs per day sleeping` ~ Gender + Year + Age + Age:Gender + Day:Gender + Year:Age + Year:Day, family = "gaussian", data = Sleep.test,
  prior_intercept = normal(0, 2.5, autoscale = TRUE),
  prior = normal(0, 2.5, autoscale = TRUE),
  prior_aux = exponential(1, autoscale = TRUE),
  chains = 4, iter = 5000*2, seed = 84735)
set.seed(84735)
```

```
set.seed(84735)
 prediction2 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 2,Gender = 2, Day = 1 ))
 write.csv(prediction2,file="Avg.sleep.hours.predictions2.csv")
 colMeans(prediction2)
#        1
#8.785299

 set.seed(84735)
 prediction3 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 3,Gender = 2, Day = 1 ))
 write.csv(prediction3,file="Avg.sleep.hours.predictions3.csv")
 colMeans(prediction3)
#        1
#8.720025

 set.seed(84735)
 prediction4 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 4, Gender = 2, Day = 1 ))
 write.csv(prediction4,file="Avg.sleep.hours.predictions4.csv")
 colMeans(prediction4)
#        1
#8.654752

 set.seed(84735)
 prediction5 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 5, Gender = 2, Day = 1 ))
 write.csv(prediction5,file="Avg.sleep.hours.predictions5.csv")
 colMeans(prediction5)
#        1
```

```
#8.589479

 set.seed(84735)
 prediction6 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 6, Gender = 2, Day = 1 ))
 write.csv(prediction6,file="Avg.sleep.hours.predictions6.csv")
 colMeans(prediction6)
#       1
#8.524205

 set.seed(84735)
 prediction7 <- posterior_predict(
  Sleep, newdata = data.frame(Year = 2017, Age = 7, Gender = 2, Day = 1 ))
 write.csv(prediction7,file="Avg.sleep.hours.predictions7.csv")
 colMeans(prediction7)
 #      1
#8.458932
```